

The Metric Cotype Gap Conjecture

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Source Problem

The source paper is:

E. Veomett and K. Wildrick, *Spaces of small metric cotype*, arXiv:1001.3326.

The source defines q_X to be the infimum of the metric cotype exponents supported by a metric space X , allowing arbitrary scaling functions $m_{p,q}(n) \in 2\mathbb{N}$. Conjecture 1.6 asks:

There is no metric space (X, d) for which $1 < q_X < 2$.

direction of a converse to Theorem 1.3.

Theorem 1.4. *Suppose that (X, d) is a 1-snowflake. Then $q_X \geq 2$.*

Theorem 1.5. *Suppose that (X, d) is a s -snowflake for some $s \in [1, \infty)$. If $1 \leq p \leq q < 2$, then (X, d) does not support a (p, q) -metric cotype inequality in which $m_{p,q}(n) = O(n^{1/q})$.*

As mentioned in [12] and [?], it is of interest to know if the conclusion of Theorem 1.5 implies that $q_X \geq 2$. Also, it would be interesting to know if the hypothesis in Theorem 1.5 can be weakened to the condition that $s_X < \infty$, and if the hypothesis in Theorem 1.3 can be weakened to the condition $s_X = \infty$.

Comparing Theorems 1.3 and 1.5 leads us to the following conjecture. Similar phenomena have been observed for conformal dimension; see [8].

Conjecture 1.6. *There is no metric space (X, d) for which $1 < q_X < 2$.*

It seems likely that a similar analysis could be made of the corresponding notion of metric type given in [11].

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2. CLASSES OF MAPPINGS

We now establish notation for the various classes of mappings we will consider throughout

Result

Theorem 1. *For every metric space (X, d) , either*

$$q_X = 1 \quad \text{or} \quad q_X \geq 2.$$

Consequently Conjecture 1.6 of Veomett–Wildrick is true.

Proof Intuition

The source paper already proves the hard lower-bound side for spaces whose snowflake index is exactly 1: such spaces have $q_X \geq 2$. Thus the only remaining possibility for a gap would be a space whose snowflake index is strictly bigger than 1.

But if d^s is a metric for some $s > 1$, then d^q is a metric for every $1 < q \leq s$. Once d^q is a metric, the metric cotype q inequality is almost just the triangle inequality on the discrete torus: a jump by $m/2$ in one coordinate is bounded by a path of $m/2$ unit jumps. The right-hand side of the metric cotype inequality averages over $\{-1, 0, 1\}^n$, so it sees a given unit coordinate jump only with probability 3^{-n} . Choosing the scaling parameter $m(n)$ exponentially large absorbs this loss. This gives metric cotype q for every $q > 1$ below the snowflake exponent, so $q_X = 1$.

Definitions

Let $1 \leq p \leq q < \infty$. A metric space (X, d) supports a (p, q) -metric cotype inequality if there are $\Gamma \geq 1$ and a scaling function $m_{p,q} : \mathbb{N} \rightarrow 2\mathbb{N}$ such that, for every n and every $f : \mathbb{Z}_m^n \rightarrow X$, with $m = m_{p,q}(n)$,

$$\mathbb{E}_{\epsilon \in \mathbb{Z}_m^n} \sum_{j=1}^n d(f(\epsilon), f(\epsilon + (m/2)e_j))^p \leq \Gamma^p m^p n^{1-p/q} \mathbb{E}_{\epsilon \in \mathbb{Z}_m^n} \mathbb{E}_{\delta \in \{-1, 0, 1\}^n} d(f(\epsilon), f(\epsilon + \delta))^p. \quad (1)$$

The exponent q_X is the infimum of the q 's for which X supports some (p, q) -metric cotype inequality with $1 \leq p \leq q$.

Following the source paper, X is an L^s -metric space if d^s is a metric, equivalently

$$d(x, y) \leq (d(x, z)^s + d(z, y)^s)^{1/s}.$$

Its snowflake index is

$$s_X = \sup\{s : X \text{ is bi-Lipschitz equivalent to an } L^s\text{-metric space}\}.$$

The Elementary Torus Lemma

Lemma 1. *Let $q > 1$. If $\rho = d^q$ is a metric on X , then (X, d) supports a (q, q) -metric cotype inequality. In particular $q_X \leq q$.*

Proof. Fix n . Choose an even integer $m = m(n)$ such that

$$m^{q-1} \geq \frac{3^n}{2}. \quad (2)$$

Let $f : \mathbb{Z}_m^n \rightarrow X$. Since $\rho = d^q$ is a metric, for each coordinate j and each $\epsilon \in \mathbb{Z}_m^n$,

$$\rho(f(\epsilon), f(\epsilon + (m/2)e_j)) \leq \sum_{k=0}^{m/2-1} \rho(f(\epsilon + ke_j), f(\epsilon + (k+1)e_j)).$$

Averaging over ϵ and using translation invariance on $\mathbb{Z}_m^n$ gives

$$\mathbb{E}_{\epsilon} \rho(f(\epsilon), f(\epsilon + (m/2)e_j)) \leq \frac{m}{2} \mathbb{E}_{\epsilon} \rho(f(\epsilon), f(\epsilon + e_j)).$$

Summing over j , set

$$S = \mathbb{E}_\epsilon \sum_{j=1}^n \rho(f(\epsilon), f(\epsilon + e_j)).$$

Then the left-hand side of (1), with $p = q$, is at most $(m/2)S$.

On the other hand, the average over $\delta \in \{-1, 0, 1\}^n$ includes each $\delta = e_j$ with probability 3^{-n} . Therefore

$$\mathbb{E}_\epsilon \mathbb{E}_\delta \rho(f(\epsilon), f(\epsilon + \delta)) \geq 3^{-n} S.$$

Combining this with (2),

$$\mathbb{E}_\epsilon \sum_{j=1}^n d(f(\epsilon), f(\epsilon + (m/2)e_j))^q \leq \frac{m}{2} S \leq m^q \mathbb{E}_\epsilon \mathbb{E}_\delta d(f(\epsilon), f(\epsilon + \delta))^q.$$

This is (1) for $p = q$, $\Gamma = 1$, and the scaling function $m(n)$. $\square$

Lemma 2. *If $s_X > 1$, then $q_X = 1$.*

Proof. Choose $s > 1$ such that X is bi-Lipschitz equivalent to an L^s -metric space (Y, ρ) . Let $1 < q \leq s$. Since ρ^s is a metric and $q/s \leq 1$, the function

$$\rho^q = (\rho^s)^{q/s}$$

is also a metric. By the preceding lemma, $q_Y \leq q$. Metric cotype is invariant under bi-Lipschitz equivalence, as recorded in Proposition 5.1 of the source paper, so $q_X \leq q$. Since this holds for every $q > 1$ sufficiently close to 1, we get $q_X = 1$. $\square$

Proof of the Gap Dichotomy

Every metric space is an L^1 -metric space, so $s_X \geq 1$. If $s_X > 1$, the preceding lemma gives $q_X = 1$. If $s_X = 1$, then X is a 1-snowflake in the terminology of the source paper. Theorem 1.4 of Veomett–Wildrick states that every 1-snowflake has $q_X \geq 2$.

Thus each metric space satisfies either $q_X = 1$ or $q_X \geq 2$. In particular no metric space satisfies $1 < q_X < 2$, proving Conjecture 1.6.

Verification Notes

The new ingredient is the elementary torus lemma. Its scaling function is exponential in n , for instance one may take any even integer

$$m(n) \geq \left(\frac{3^n}{2} \right)^{1/(q-1)}.$$

This is essential: the source paper’s Theorem 1.5 rules out $q < 2$ for finite-snowflake-index spaces only when the scaling function has sharp growth $m_{p,q}(n) = O(n^{1/q})$. The proof here uses the freedom in the definition of q_X , where arbitrary scaling functions are allowed.

No computation is used. A human reviewer should check the convention that q_X in the source paper allows arbitrary scaling functions, and the definition of s_X that makes $s_X = 1$ exactly the 1-snowflake case.

Novelty Check

On 2026-06-28, I checked the lightweight run indexes for arXiv:1001.3326, “metric cotype”, “small metric cotype”, “ q_X ”, and “snowflake index”. There were nearby packets about metric cotype, but no exact packet for Conjecture 1.6 or the q_X -gap dichotomy.

I also searched the web for exact and close variants: “Spaces of small metric cotype Conjecture 1.6”, “ q_X s X metric cotype”, “metric cotype snowflake index q_X ”, and “metric cotype d_q is a metric”. The bounded search found the source paper but no later explicit solution of the conjecture.

Human Review Recommendation

This is a candidate full solution. The main review point is whether the definition of q_X indeed permits the exponential scaling function used in the torus lemma. Under the source paper’s Definition 1.1 and the displayed definition of q_X , it does.

References

References

- [1] E. Veomett and K. Wildrick, *Spaces of small metric cotype*, arXiv:1001.3326.